

[OK] Dataset final cargado exitosamente.
El dataset tiene 288 filas y 10 columnas.

--- Primeras 5 filas del Dataset ---

	provincia	latitude	longitude	month	temp_avg	wind_speed_avg	\
0	Azuay	-2.9006	-79.004	1	11.5773	1.8933	
1	Azuay	-2.9006	-79.004	2	12.0730	1.5313	
2	Azuay	-2.9006	-79.004	3	12.3677	1.4574	
3	Azuay	-2.9006	-79.004	4	11.8671	1.6135	
4	Azuay	-2.9006	-79.004	5	11.5593	1.8098	

	humidity_avg	target_solar_kwh_per_m2	target_wind_mwh	target_hydro_mwh
0	88.3806	18.7648	53.6837	6852.7414
1	87.4912	17.1785	25.5508	8311.9789
2	86.3080	21.4602	24.5915	11468.7961
3	88.6220	18.1111	32.2810	8469.2523
4	85.7130	20.9756	42.0964	5323.7848

--- Informacion general del Dataset (df.info()) ---

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 288 entries, 0 to 287  
Data columns (total 10 columns):  
#   Column                                Non-Null Count  Dtype  
---  -  
0   provincia                             288 non-null    object  
1   latitude                             288 non-null    float64  
2   longitude                             288 non-null    float64  
3   month                                288 non-null    int64  
4   temp_avg                             288 non-null    float64  
5   wind_speed_avg                       288 non-null    float64  
6   humidity_avg                         288 non-null    float64  
7   target_solar_kwh_per_m2              288 non-null    float64  
8   target_wind_mwh                      288 non-null    float64  
9   target_hydro_mwh                    288 non-null    float64  
dtypes: float64(8), int64(1), object(1)  
memory usage: 22.6+ KB
```

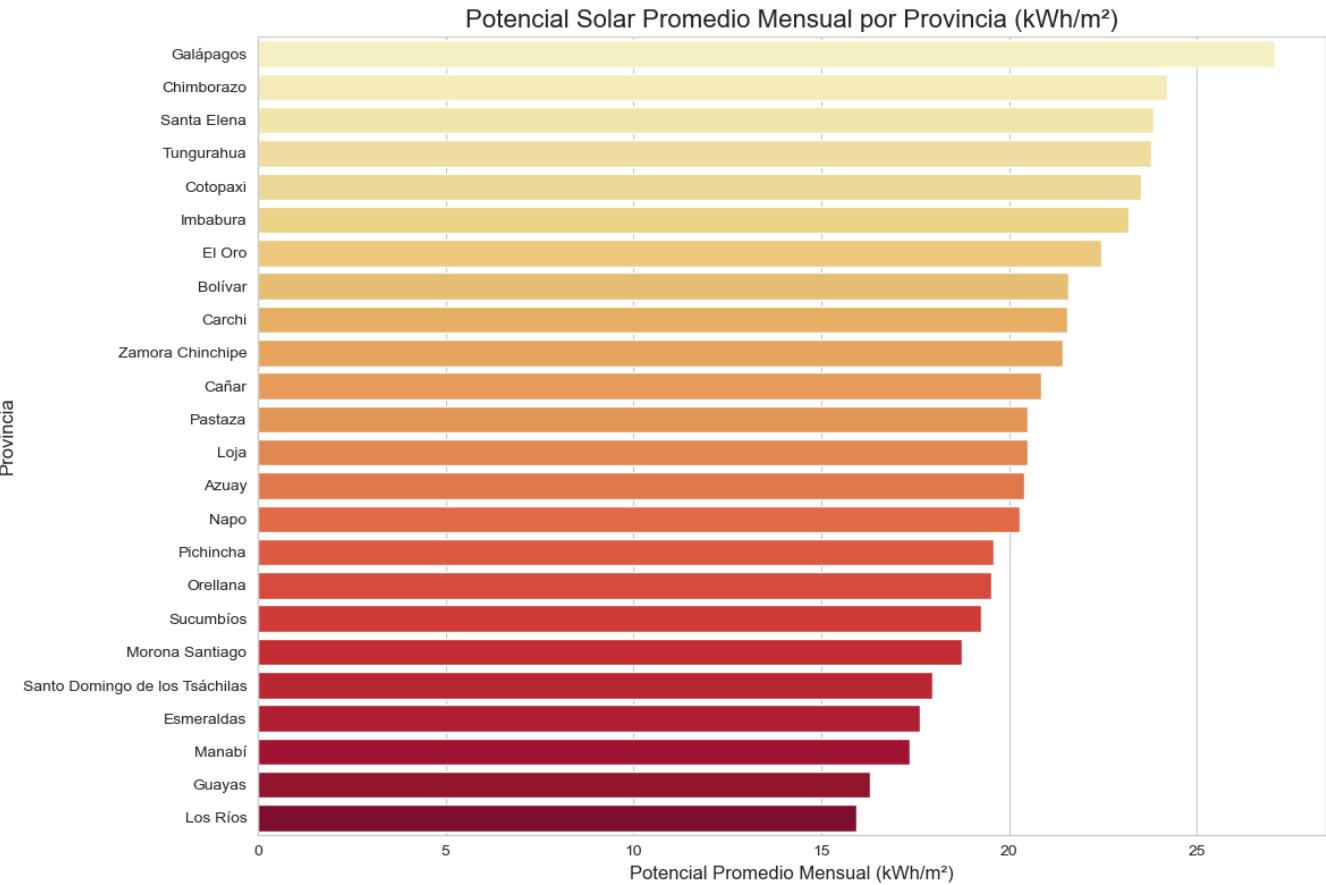
--- Estadisticas descriptivas del Dataset ---

	latitude	longitude	month	temp_avg	wind_speed_avg	\
count	288.000000	288.000000	288.000000	288.000000	288.000000	
mean	-1.442762	-79.260017	6.500000	17.869299	1.405835	
std	1.405490	2.359823	3.458061	5.612944	1.294411	
min	-4.068900	-89.616000	1.000000	7.951200	0.546800	
25%	-2.247500	-79.549600	3.750000	12.268100	0.876075	
50%	-1.365850	-78.974800	6.500000	19.135950	1.057200	
75%	-0.282700	-78.120900	9.250000	23.327275	1.474725	
max	0.966100	-76.896900	12.000000	26.362100	16.757800	

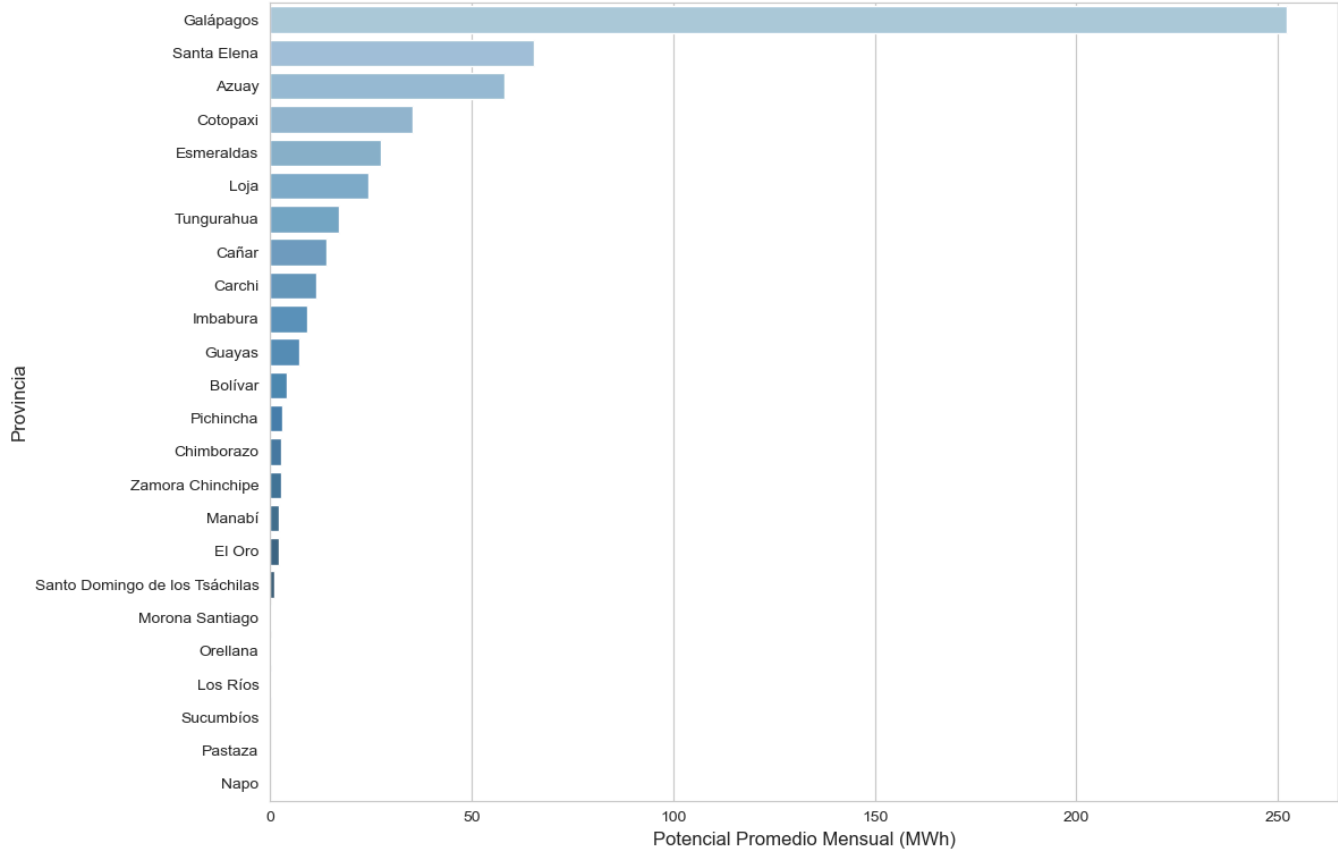
	humidity_avg	target_solar_kwh_per_m2	target_wind_mwh	\
count	288.000000	288.000000	288.000000	
mean	84.444571	20.726006	22.589086	
std	5.573086	3.417101	60.083632	
min	61.287600	13.477800	0.000000	
25%	81.717725	18.217000	0.243200	

50%	85.549700	20.382750	3.756250
75%	88.043625	23.111875	16.114575
max	94.008900	31.947100	450.198700

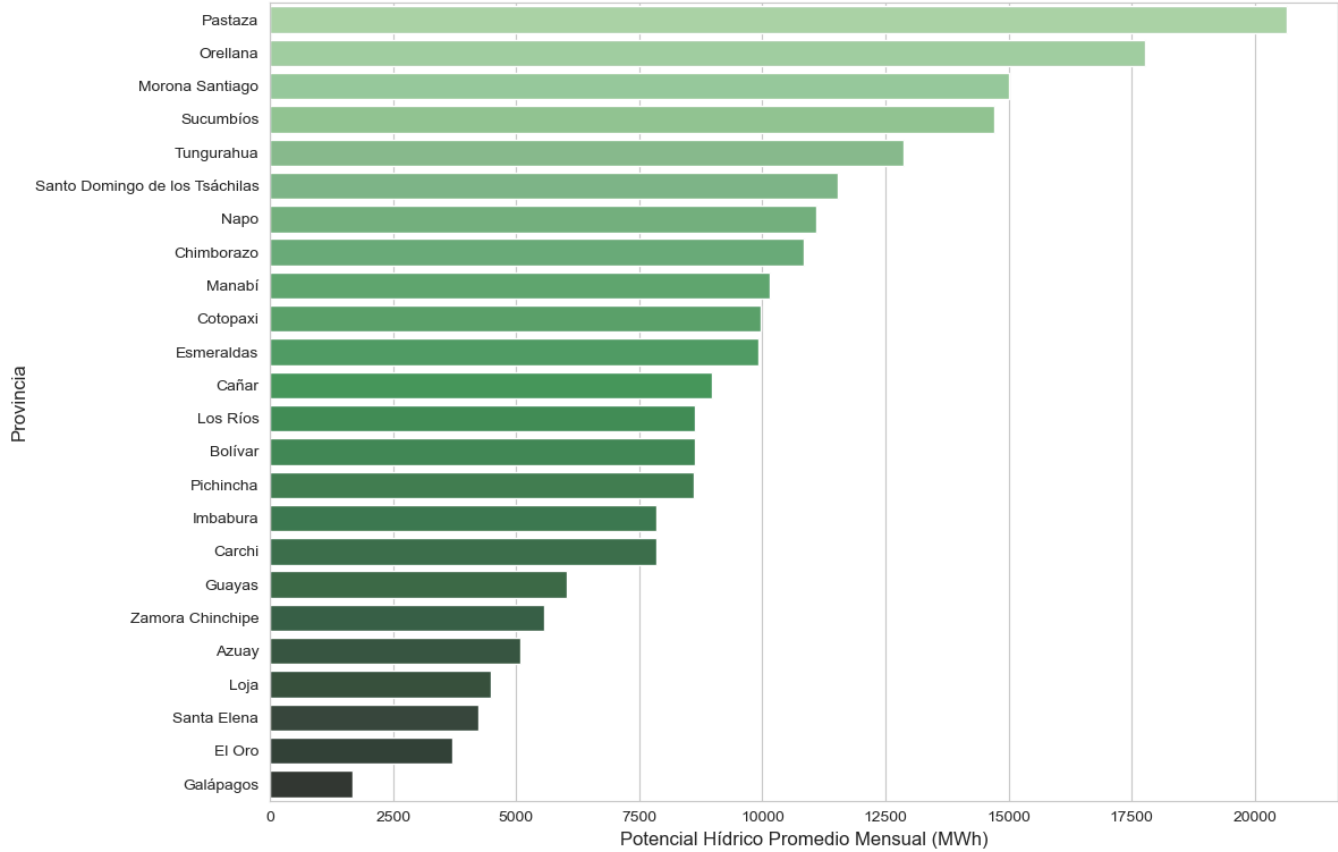
	target_hydro_mwh
count	288.000000
mean	9412.526624
std	6428.176094
min	62.422900
25%	3548.379200
50%	8651.251650
75%	13851.203975
max	27390.698600



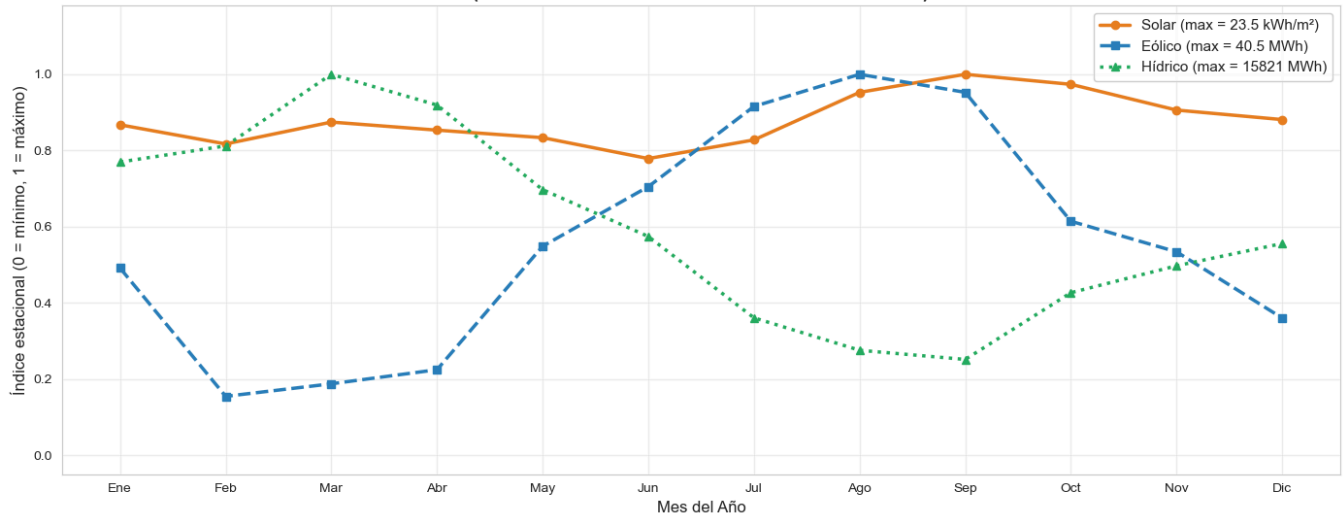
Potencial Eólico Promedio Mensual por Provincia (MWh)



Potencial Hídrico Promedio Mensual por Provincia (MWh)



Estacionalidad Relativa del Potencial Energético en Ecuador
(Normalizada al máximo anual de cada recurso)



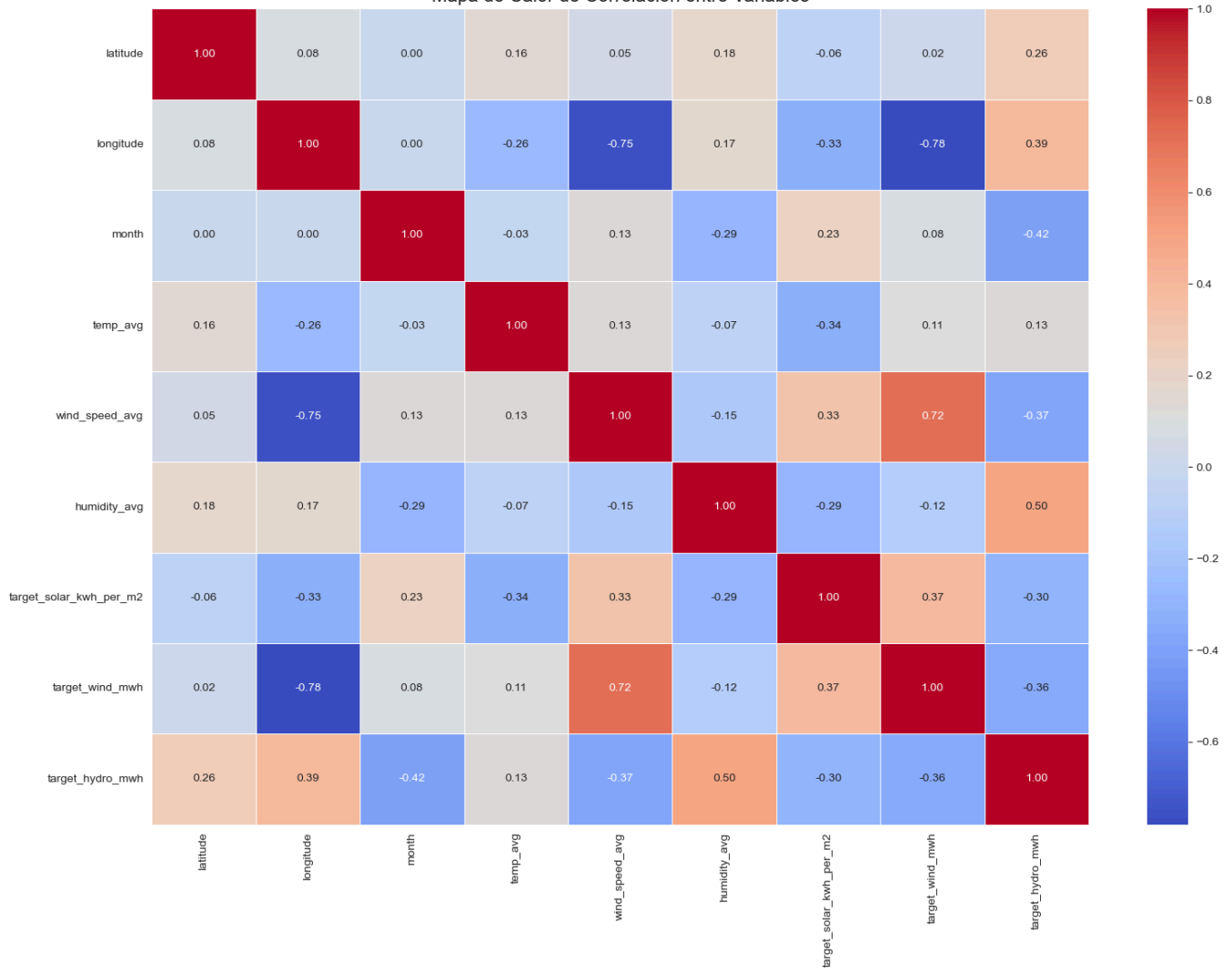
[OK] Estacionalidad normalizada guardada: docs/estacionalidad.png

Solar: pico en mes 9 (23.54 kWh/m²)

Eolico: pico en mes 8 (40.53 MWh)

Hidrico: pico en mes 3 (15820.87 MWh)

Mapa de Calor de Correlación entre Variables



--- Correlación con el Potencial Solar ---

```
target_solar_kwh_per_m2    1.000000
target_wind_mwh            0.374743
wind_speed_avg             0.330725
month                     0.231864
latitude                  -0.064211
humidity_avg              -0.286569
target_hydro_mwh          -0.304156
longitude                 -0.328059
temp_avg                  -0.341202
Name: target_solar_kwh_per_m2, dtype: float64
```

--- Correlación con el Potencial Eólico ---

```
target_wind_mwh            1.000000
wind_speed_avg             0.721760
target_solar_kwh_per_m2    0.374743
temp_avg                   0.109186
month                     0.082917
latitude                   0.017904
humidity_avg              -0.123951
target_hydro_mwh          -0.355435
longitude                 -0.780134
Name: target_wind_mwh, dtype: float64
```

--- Correlación con el Potencial Hídrico ---

```
target_hydro_mwh           1.000000
humidity_avg               0.502006
longitude                  0.390367
latitude                   0.264417
temp_avg                   0.129329
target_solar_kwh_per_m2   -0.304156
target_wind_mwh           -0.355435
wind_speed_avg            -0.373304
month                     -0.417220
Name: target_hydro_mwh, dtype: float64
```

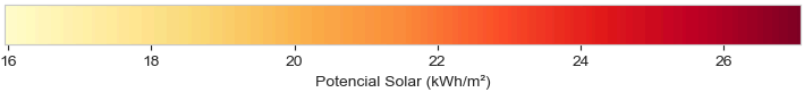
[OK] GeoJSON local cargado: 24 provincias

Provincias con datos en el mapa: 24/24

Diagnostico Galapagos:

	nombre	target_solar_kwh_per_m2	target_wind_mwh	target_hydro_mwh
19	Galápagos	27.077725	252.195183	1685.284208

Potencial Solar Promedio Mensual por Provincia en Ecuador



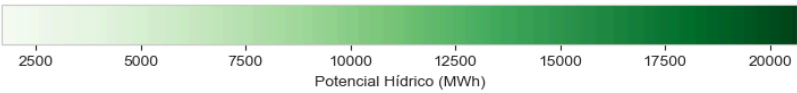
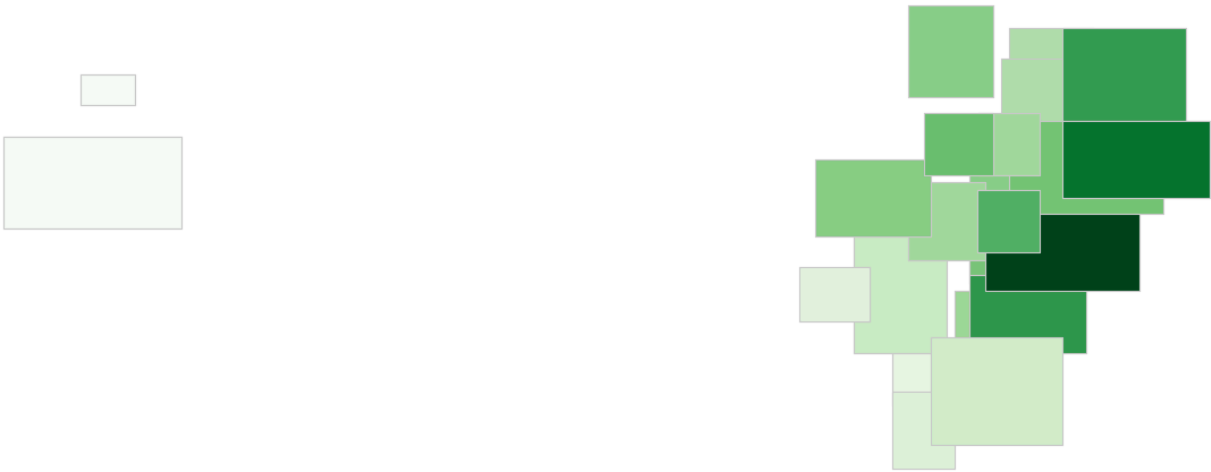
[OK] Mapa solar guardado

Potencial Eólico Promedio Mensual por Provincia en Ecuador



[OK] Mapa eolico guardado

Potencial Hídrico Promedio Mensual por Provincia en Ecuador



[OK] Mapa hidrico guardado

[OK] Shapefile cargado.

--- Columnas encontradas en el Shapefile ---

['featurecla', 'scalerank', 'adm1_code', 'diss_me', 'iso_3166_2', 'wikipedia', 'iso_a2', 'adm0_sr', 'name', 'name_alt', 'name_local', 'type', 'type_en', 'code_local', 'code_hasc', 'note', 'hasc_maybe', 'region', 'region_cod', 'provnum_ne', 'gadm_level', 'check_me', 'datarank', 'abbrev', 'postal', 'area_sqkm', 'sameascity', 'labelrank', 'name_len', 'mapcolor9', 'mapcolor13', 'fips', 'fips_alt', 'woe_id', 'woe_label', 'woe_name', 'latitude', 'longitude', 'sov_a3', 'adm0_a3', 'adm0_label', 'admin', 'geonunit', 'gu_a3', 'gn_id', 'gn_name', 'gns_id', 'gns_name', 'gn_level', 'gn_region', 'gn_a1_code', 'region_sub', 'sub_code', 'gns_level', 'gns_lang', 'gns_adm1', 'gns_region', 'min_label', 'max_label', 'min_zoom', 'wikidataid', 'name_ar', 'name_bn', 'name_de', 'name_en', 'name_es', 'name_fr', 'name_el', 'name_hi', 'name_hu', 'name_id', 'name_it', 'name_ja', 'name_ko', 'name_nl', 'name_pl', 'name_pt', 'name_ru', 'name_sv', 'name_tr', 'name_vi', 'name_zh', 'ne_id', 'name_he', 'name_uk', 'name_ur', 'name_fa', 'name_zht', 'FCLASS_ISO', 'FCLASS_US', 'FCLASS_FR', 'FCLASS_RU', 'FCLASS_ES', 'FCLASS_CN', 'FCLASS_TW', 'FCLASS_IN', 'FCLASS_NP', 'FCLASS_PK', 'FCLASS_DE', 'FCLASS_GB', 'FCLASS_BR', 'FCLASS_IL', 'FCLASS_PS', 'FCLASS_SA', 'FCLASS_EG', 'FCLASS_MA', 'FCLASS_PT', 'FCLASS_AR', 'FCLASS_JP', 'FCLASS_KO', 'FCLASS_VN', 'FCLASS_TR', 'FCLASS_ID', 'FCLASS_PL', 'FCLASS_GR', 'FCLASS_IT', 'FCLASS_NL', 'FCLASS_SE', 'FCLASS_BD', 'FCLASS_UA', 'FCLASS_TLC', 'geometry']

--- Valores unicos en la columna 'admin' ---

['Argentina' 'Uruguay' 'Indonesia' 'Malaysia' 'Chile' 'Bolivia' 'Peru' 'Dhekelia Sovereign Base Area' 'Cyprus' 'India' 'China' 'Israel' 'Gaza Strip' 'West Bank' 'Lebanon' 'Ethiopia' 'S. Sudan' 'Somalia' 'Kenya' 'Malawi' 'United Republic of Tanzania' 'Syria' 'Somaliland' 'France' 'Suriname' 'Guyana' 'South Korea' 'North Korea' 'Morocco' 'Western Sahara' 'Costa Rica' 'Nicaragua' 'Republic of the Congo' 'Democratic Republic of the Congo' 'Bhutan' 'Ukraine' 'Belarus' 'Namibia' 'South Africa' 'Saint Martin' 'Sint Maarten' 'United Arab Emirates' 'Oman' 'Uzbekistan' 'Kazakhstan' 'Tajikistan' 'Lithuania' 'Brazil' 'Mongolia' 'Russia' 'Czech Republic' 'Germany' 'Estonia' 'Latvia' 'Norway' 'Sweden' 'Finland' 'Vietnam' 'Cambodia' 'Luxembourg' 'Belgium' 'Georgia' 'Macedonia' 'Albania' 'Azerbaijan' 'Kosovo' 'Turkey' 'Spain' 'Laos' 'Kyrgyzstan' 'Armenia' 'Denmark' 'Libya' 'Tunisia' 'Romania' 'Hungary' 'Slovakia' 'Poland' 'Ireland' 'United Kingdom' 'Greece' 'Zambia' 'Sierra Leone' 'Guinea' 'Liberia' 'Central African Republic' 'Sudan' 'Djibouti' 'Eritrea' 'Austria' 'Iraq' 'Italy' 'Switzerland' 'Iran' 'Netherlands' 'Liechtenstein' 'Ivory Coast' 'Republic of Serbia' 'Mali' 'Senegal' 'Nigeria' 'Benin' 'Angola' 'Croatia' 'Slovenia' 'Qatar' 'Saudi Arabia' 'Botswana' 'Zimbabwe' 'Pakistan' 'Bulgaria' 'Thailand' 'San Marino' 'Haiti' 'Dominican Republic' 'Chad' 'Kuwait' 'El Salvador' 'Guatemala' 'East Timor' 'Brunei' 'Monaco' 'Algeria' 'Mozambique' 'Swaziland' 'Burundi' 'Rwanda' 'Myanmar' 'Bangladesh' 'Andorra' 'Afghanistan' 'Montenegro' 'Bosnia and Herzegovina' 'Uganda' 'US Naval Base Guantanamo Bay' 'Cuba' 'Honduras' 'Ecuador' 'Colombia' 'Paraguay' 'Portugal' 'Moldova' 'Turkmenistan' 'Jordan' 'Nepal' 'Lesotho' 'Cameroon' 'Gabon' 'Niger' 'Burkina Faso' 'Togo' 'Ghana' 'Guinea Bissau' 'Gibraltar' 'United States of America' 'Canada' 'Mexico' 'Belize' 'Panama' 'Venezuela' 'Papua New Guinea' 'Egypt' 'Yemen' 'Mauritania' 'Equatorial Guinea' 'Gambia' 'Hong Kong S.A.R.' 'Vatican' 'Northern Cyprus' 'Siachen Glacier' 'Baykonur Cosmodrome' 'Akrotiri Sovereign Base Area' 'Antarctica' 'New Zealand' 'New Caledonia' 'Australia' 'Madagascar' 'Philippines' 'Sri Lanka' 'Curaçao' 'Aruba']

'The Bahamas' 'Turks and Caicos Islands' 'Taiwan' 'Japan'
'Saint Pierre and Miquelon' 'Iceland' 'Greenland' 'Pitcairn Islands'
'French Polynesia' 'French Southern and Antarctic Lands' 'Seychelles'
'Kiribati' 'Marshall Islands' 'Trinidad and Tobago' 'Grenada'
'Caribbean Netherlands' 'Saint Vincent and the Grenadines' 'Barbados'
'Saint Lucia' 'Dominica' 'United States Minor Outlying Islands'
'Montserrat' 'Antigua and Barbuda' 'Saint Kitts and Nevis'
'United States Virgin Islands' 'Saint Barthelemy' 'Puerto Rico'
'Anguilla' 'British Virgin Islands' 'Jamaica' 'Cayman Islands' 'Bermuda'
'Heard Island and McDonald Islands' 'Saint Helena' 'Mauritius' 'Comoros'
'Sao Tome and Principe' 'Cape Verde' 'Malta' 'Jersey' 'Guernsey'
'Isle of Man' 'Aland' 'Faroe Islands' 'Indian Ocean Territories'
'British Indian Ocean Territory' 'Singapore' 'Norfolk Island'
'Cook Islands' 'Fiji' 'Tonga' 'Wallis and Futuna' 'Samoa'
'Solomon Islands' 'Tuvalu' 'Maldives' 'Nauru'
'Federated States of Micronesia' 'South Georgia and the Islands'
'Falkland Islands' 'Vanuatu' 'Niue' 'American Samoa' 'Palau' 'Guam'
'Northern Mariana Islands' 'Bahrain' 'Coral Sea Islands' 'Spratly Is.'
'Clipperton Island' 'Macau S.A.R' 'Ashmore and Cartier Islands']

--- Valores unicos en la columna 'name' (primeros 10) ---

['Entre Ríos' 'Paysandú' 'Kalimantan Timur' 'Sabah' 'Arica y Parinacota'
'La Paz' 'Oruro' 'Tarapacá' 'Potosí' 'Antofagasta']